

Conclusion - We had studied about the functional unit of GSM & also signaling technique of it.



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Experiment No. 3

Title:- To Study of GPRS Services

Problem Statement:- To study of GPRS services

Objective:- To understand the GPRS Protocol.

Theory:-

GPRS is an expansion global system for Mobile Communication. It is a packet-oriented mobile data standard on 2G and 3G cellular communication networks global system for mobile communication. GPRS was built up by European Telecommunications Standard Institute (ETSI) because of prior GPRS, and I-mode packet switched cell advances.

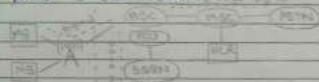
GPRS overrides the wired associations, as this framework has streamlined access to packet info network like the web. The packet radio standard is utilized by GPRS to transport client info. packets in structured route between GSM versatile stations and external packet information networks. These packets can be straightforwardly directed to packet changed system from GPRS portable stations.

In GSM architecture there is one component called BSC. But in GPRS there is one component is added to BSC called PCU. PCU stands for Packet Control Unit. If signal comes to BSC and that signal contains data, then PCU routes to SGSN. Interference is used between BSC and PCU is FR1



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interference. After signal comes to SGSN it delivers the data packet to GGSN. GGSN routes the data packet to data network.

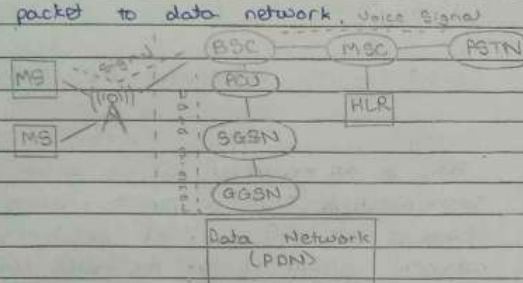


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BSS for GPRS requires installation of one or more PCU as well as software update. Moreover, the BSC may necessitate a software upgrade, but normally does not necessitate hardware requirements. When voice or data traffic originates from a mobile device, it is carried via air to BSC & then from BSC to GGSN in same manner as regular GSM voice.



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In GSM architecture there is one component called BSC. But in GPRS there is one component is added to BSC called PCU. PCU stands for Packet Control Unit. If signal comes to BSC and that signal contains data, then PCU routes to SGSN. Interface is used between BSC and PCU is FR1 interface. After signal comes to SGSN, it delivers the data packet to GGSN. It routes data packet to data Network.

GPRS Mobile Station are necessary to use GPRS service because existing GSM phones cannot process the improved air interface or packet data. MS may include a high-speed version of current phones to provide high-speed data access, a new PDA device with an in-built GSM phone & PC cards for laptop computers. These MS are backwards compatible for making GSM based voice calls.



BSS for GPRS requires installation of one or more PCU as well as software update. Moreover, the BTS may necessitate a software upgrade, but normally does not necessitate hardware requirements. When voice or data traffic originates from a mobile device, it is carried via air to BTS & then from BTS - BSC in same manner as regular GSM. However, at the output of BSC the traffic is divided; voice is transferred to MSC while data is sent to a new device called SGSN via PCU over a Frame Relay interface.

History of GPRS :-

GPRS was one of the main advances that empowered a cell system with IP systems in mid-2000s. The capacity to pursue web from a telephone whenever through "dependably on" data networking, while underestimated in a great part of world today. Instead, even now, GPRS keeps on being utilized in parts of world where it has been too expensive even to consider upgrading cell organization framework to move up to newer alternatives.

Goals of GPRS: a) Consistent IP services
b) Leverage industry investment in IP
c) Open Architecture
d) Service innovation independent of infrastructure.



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Services offered : a) SMS messaging and broadcasting.
b) Push-to-talk over cellular.
c) Instant messaging and presence.
d) Multimedia messaging service.
e) Point-to-point & Point-to-Multipoint service.

Protocols supported : a) Internet Protocol (IP)
b) Point-to-Point Protocol (PPP)

Benefits of GPRS:

- a) Mobility - Capacity to keep up consistent voice and information interchanges while moving.
- b) Cost-Efficient - Communication via GPRS is cheaper than through the regular GSM Network.
- c) Immediacy - Allows customers to obtain connectivity when needed, regardless of location, ^{and} without a lengthy login session.
- d) Localization - Enables customers to acquire data applicable to their present area.
- e) Easy Billing - GPRS packet transmission offers an easier to use billing than that offered by circuit switched administrations.

FAQ

Q1: What is GPRS?

Ans: GPRS is not completely separate network to GSM. Many of the device such as the BTS & they are still used. There are two non-functional elements which play a major role in how GPRS works. The